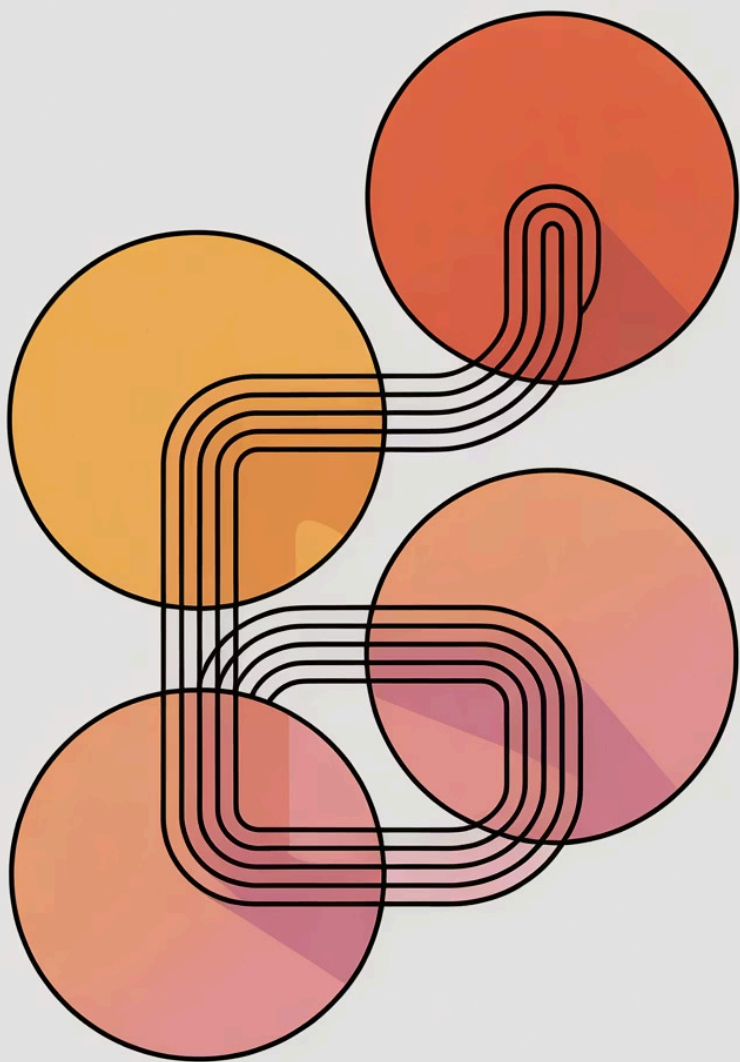


An abstract graphic on the left side of the slide. It features a background of orange and yellow wavy, organic shapes. Overlaid on these are several sets of concentric circles in a light pink or beige color, some of which are partially obscured by the larger shapes.

Understanding Naive Bayes Algorithm: From Conditional Probability to Classification

A comprehensive journey through the mathematical foundations and practical applications of one of machine learning's most elegant algorithms



Chapter 1

Foundations

What is Conditional Probability?

What is Conditional Probability?



Core Concept

The probability of event A occurring given that event B has already occurred



Mathematical Notation

$P(A|B)$ reads as "Probability of A given B"



Intuitive Example

What's the probability of drawing a queen if you already know the card is a club?

Conditional probability helps us update our beliefs about events when we gain new information about related events.

Example: Drawing Cards Without Replacement



01

Initial Setup

A standard deck contains 52 cards with 4 queens total

02

First Draw

$P(\text{Queen on 1st draw}) = 4/52 = 1/13$

03

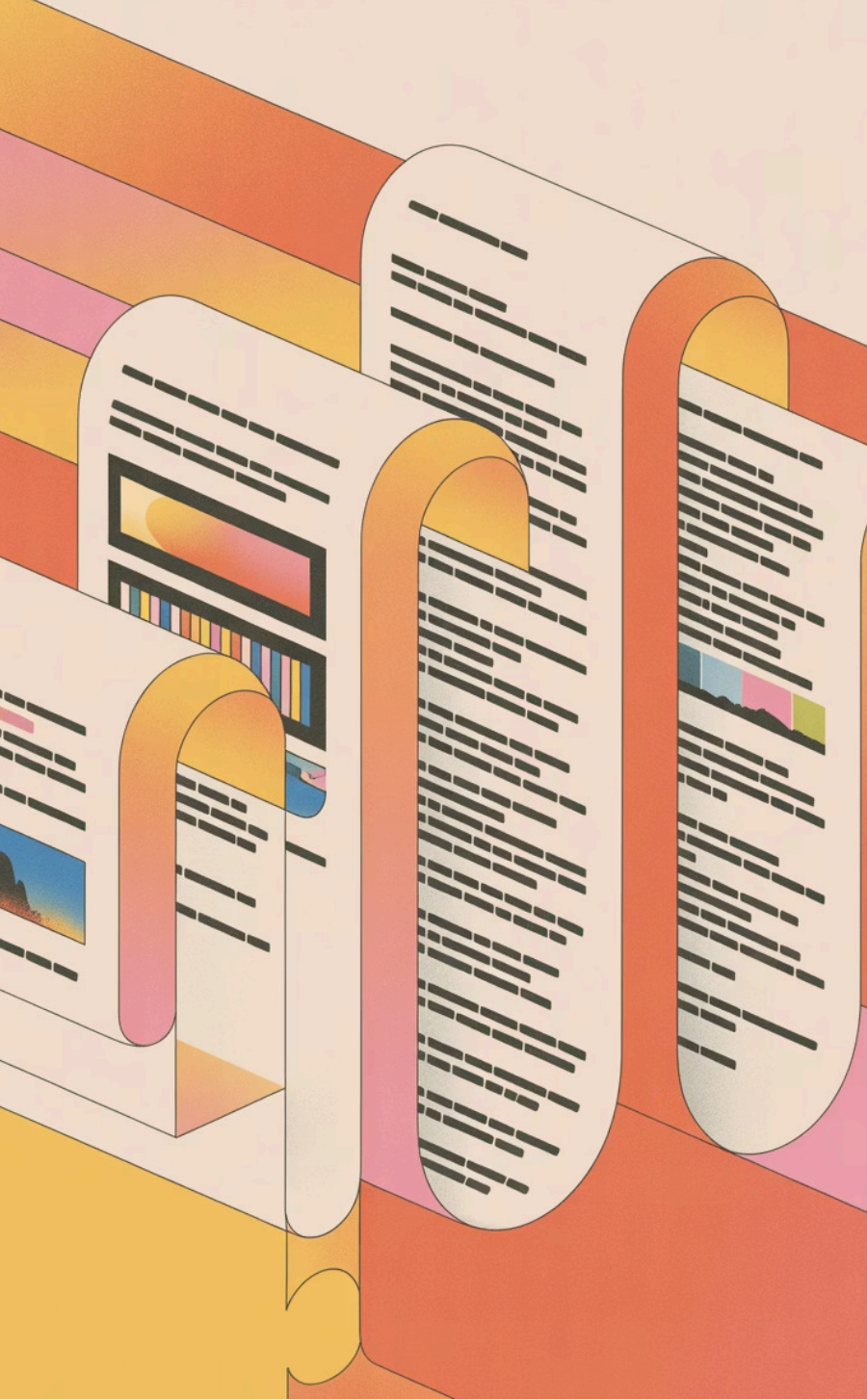
Conditional Update

If first card is a queen: $P(\text{Queen on 2nd draw}) = 3/51$

04

Key Insight

Probabilities dynamically update based on prior events



Example: Document Classification Setup

Sample Document

Document 1: Contains words w_1, w_2, w_3

Known label: 0 (negative/no)

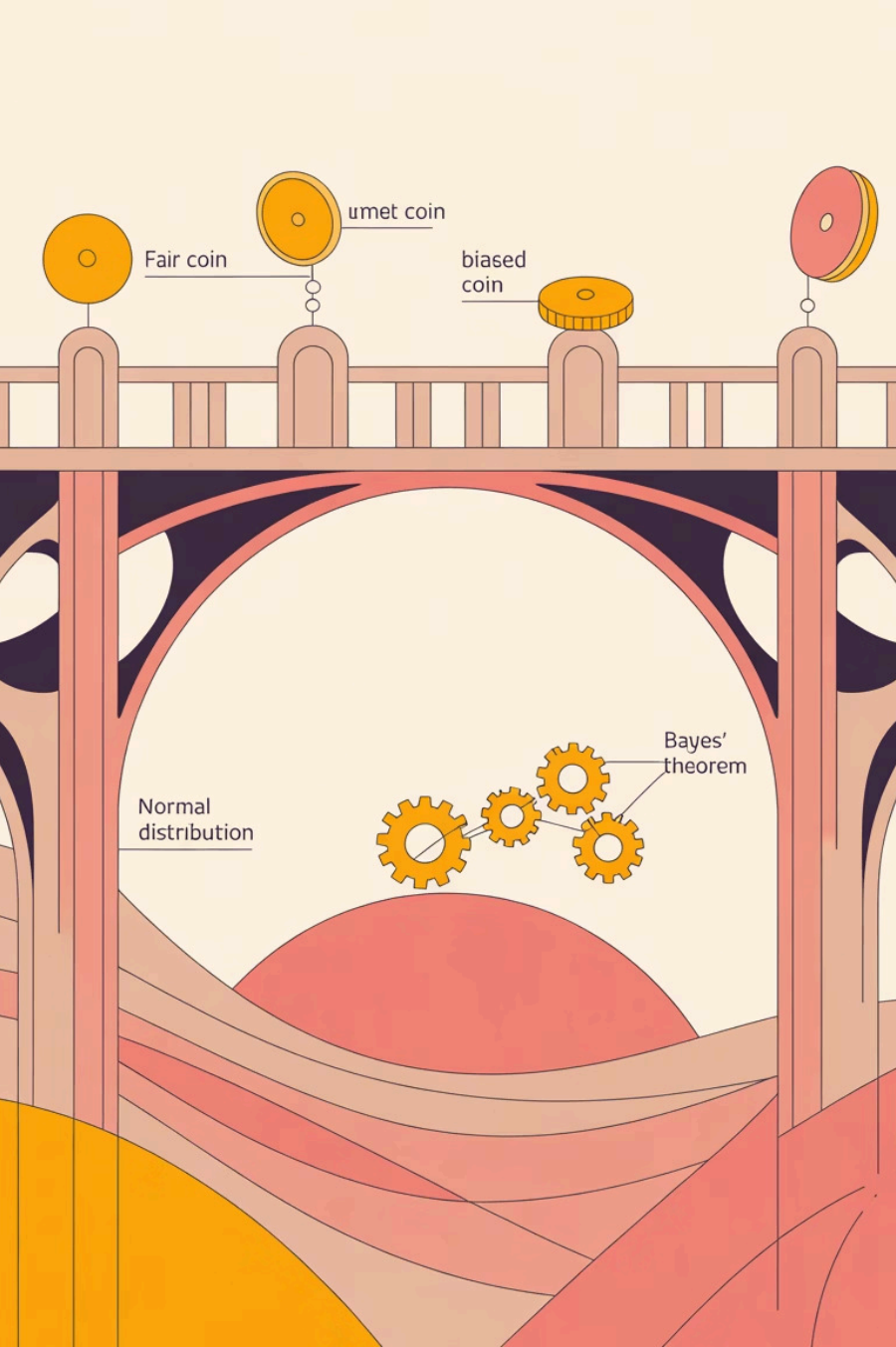
Classification Question

Given these specific words appear in a document, what is the probability the label is positive (1)?

Mathematical Goal

Calculate $P(\text{label}=1 \mid \text{words})$
using conditional probability principles

This foundational example demonstrates how we can use word patterns to predict document categories—the core principle behind text classification.



Chapter 2

Bayes' Theorem

The Bridge to Naive Bayes

Bayes' Theorem Formula

$$P(A|B) = \frac{P(B|A) \times P(A)}{P(B)}$$



P(A): Prior Probability

Our initial belief about event A before observing any evidence



P(B|A): Likelihood

Probability of observing evidence B given that A is true



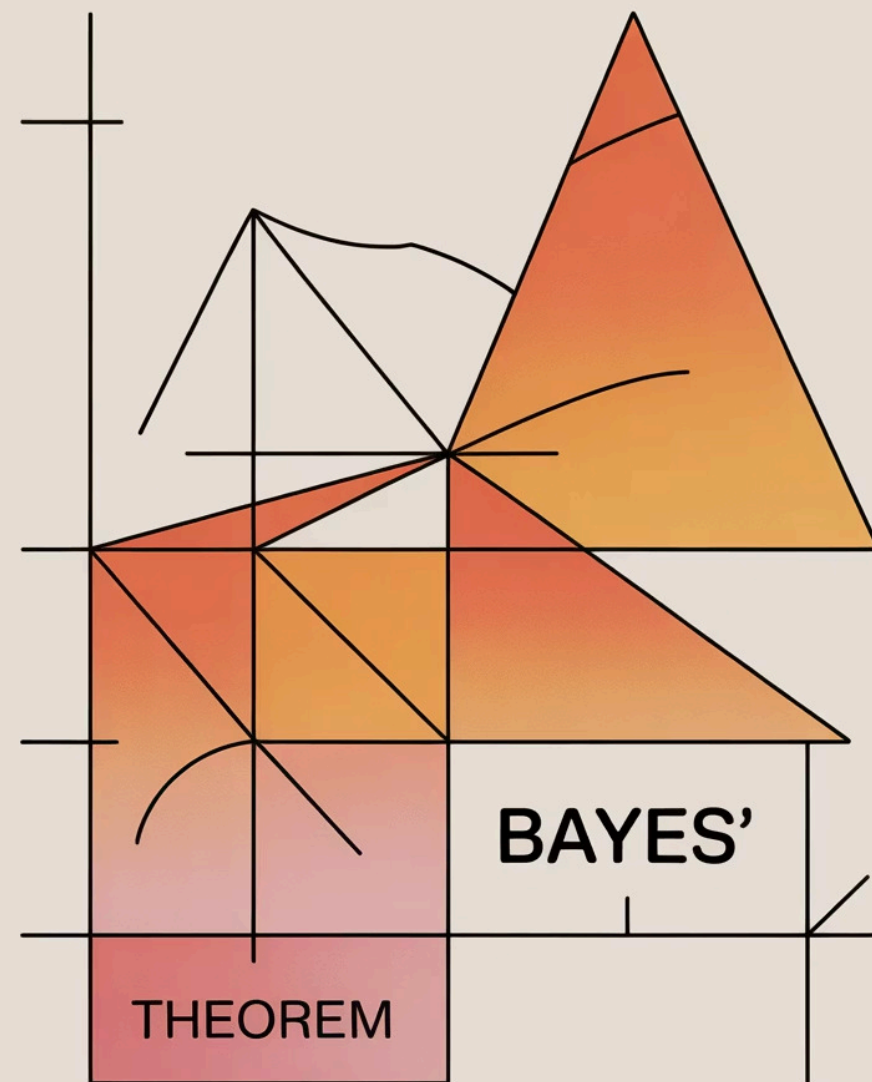
P(B): Evidence

Total probability of observing B across all scenarios



Power of Inversion

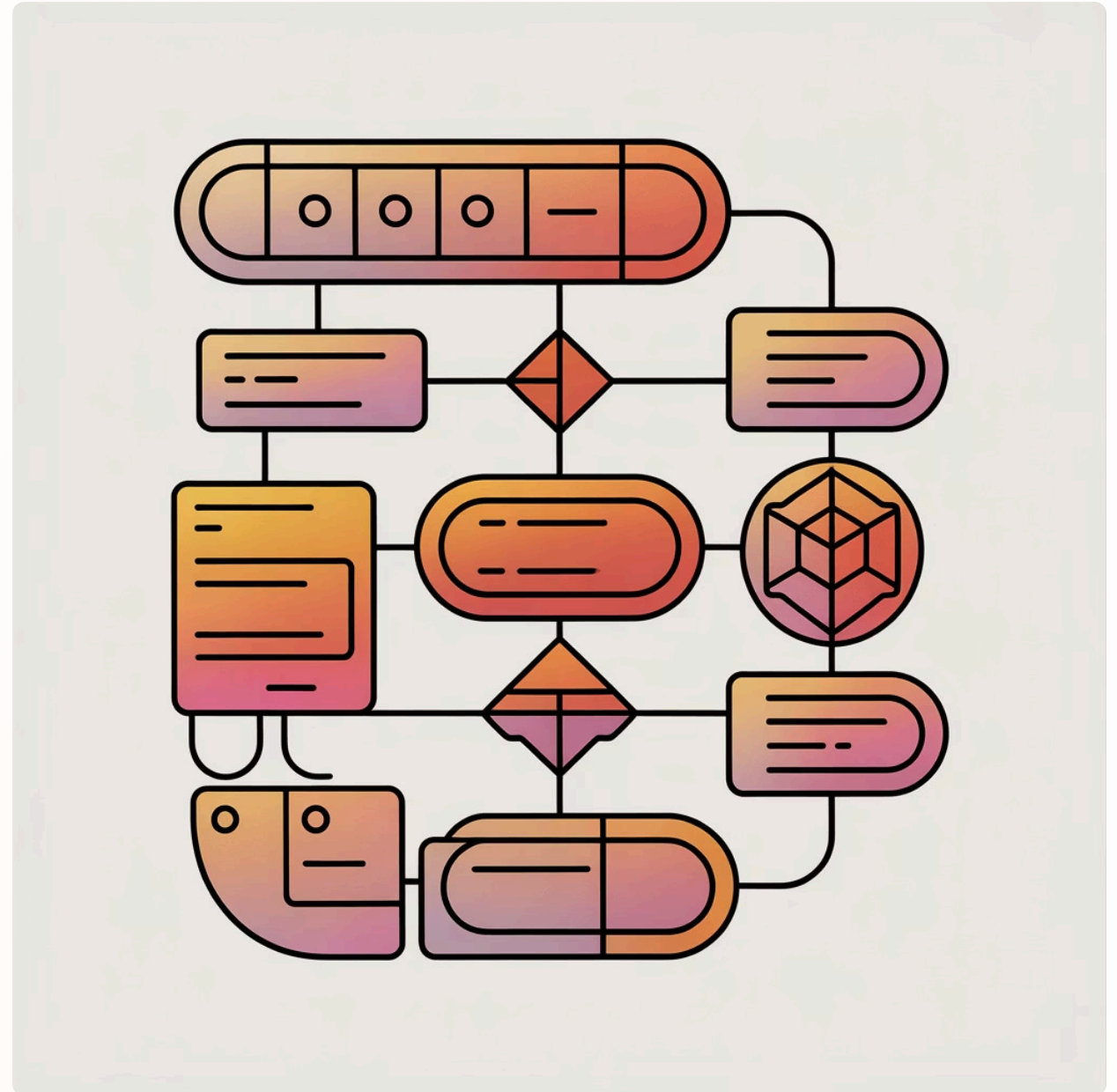
Allows reversing conditional probabilities to answer inverse questions



Applying Bayes' Theorem to Classification

The Translation

- **A = class label** (positive or negative sentiment)
- **B = observed features** (words appearing in document)
- **Goal:** Compute $P(\text{class} \mid \text{features})$



📌 **Key Insight:** By applying Bayes' Theorem, we transform the difficult question "What class does this document belong to?" into calculable probabilities based on word frequencies in our training data.

Chapter 3

Naive Bayes Algorithm

The "Naive" Assumption



What is Naive Bayes?



Probabilistic Classifier

Built directly on Bayes' theorem for classification tasks



Independence Assumption

Assumes features are conditionally independent given the class



Simplified Math

Dramatically reduces computational complexity



Practical Success

Works remarkably well despite the "naive" simplification

$$P(x_1, x_2, \dots, x_n | y) = \prod_{i=1}^n P(x_i | y)$$



Mathematical Formulation of Naive Bayes

The Classification Formula

$$P(y|X) \propto P(y) \times \prod_{i=1}^n P(x_i|y)$$



y: Class Label

The category we're trying to predict
(e.g., spam/not spam,
positive/negative)



X: Feature Vector

$X = (x_1, x_2, \dots, x_n)$ represents all observed
features like words in a document



Decision Rule

Choose class y that maximizes this
posterior probability

Detailed Example: Calculating Probabilities Step-by-Step

Our Training Data:

- Document 1 (Class 0 - Negative): "w1 w2 w2"
- Document 2 (Class 1 - Positive): "w2 w3"
- Vocabulary: {w1, w2, w3}

Word Counts by Class:

Class 0 counts:

- w1: 1 time
- w2: 2 times
- w3: 0 times
- Total words: 3

Class 1 counts:

- w1: 0 times
- w2: 1 time
- w3: 1 time
- Total words: 2

Prior Probabilities:

- $P(\text{Class 0}) = 1/2 = 0.5$ (1 document out of 2)
- $P(\text{Class 1}) = 1/2 = 0.5$ (1 document out of 2)

Calculating $P(\text{word}|\text{class})$ with Laplace Smoothing:

Formula: $P(\text{word}|\text{class}) = (\text{count}(\text{word}, \text{class}) + 1) / (\text{total words in class} + \text{vocabulary size})$

For Class 0:

- $P(w1|0) = (1+1)/(3+3) = 2/6 = 0.333$
- $P(w2|0) = (2+1)/(3+3) = 3/6 = 0.500$
- $P(w3|0) = (0+1)/(3+3) = 1/6 = 0.167$

For Class 1:

- $P(w1|1) = (0+1)/(2+3) = 1/5 = 0.200$
- $P(w2|1) = (1+1)/(2+3) = 2/5 = 0.400$
- $P(w3|1) = (1+1)/(2+3) = 2/5 = 0.400$

📌 Why Laplace Smoothing is Critical:

Without smoothing, if a word never appears in a class, its probability is 0. Since we multiply probabilities, ONE zero makes the ENTIRE product zero, destroying our ability to classify.

Example: w3 never appears in Class 0. Without smoothing:

- $P(w3|0) = 0/3 = 0$

If we need to classify a document containing w3:

$$P(\text{Class 0} | \text{document}) \propto P(0) \times \dots \times P(w3|0) \times \dots = \dots \times 0 \times \dots = \mathbf{0}$$

This makes classification impossible! Laplace smoothing adds 1 to all counts (numerator) and vocabulary size to total (denominator), ensuring no probability is ever zero while minimally affecting other probabilities.

Naive Bayes: Worked Example for Classification

Let's walk through a complete numerical example to classify a new document using the Naive Bayes algorithm.

Step 1: Define the Classification Problem

We have a **new document** to classify: "w1 w2 w2".

Our goal is to calculate the posterior probabilities $P(\text{Class } 0 \mid w_1, w_2, w_2)$ and $P(\text{Class } 1 \mid w_1, w_2, w_2)$ to determine which class the document most likely belongs to.

Step 2: Recall Training Data and Prior Probabilities

For this example, we assume our model was trained on the following data:

- Class 0 (Negative):** Training documents collectively contained words: w1, w2, w2 (total 3 words).
- Class 1 (Positive):** Training documents collectively contained words: w2, w3 (total 2 words).
- Vocabulary Size:** 3 unique words (w1, w2, w3).
- Prior Probabilities** (assuming equal class distribution): $P(\text{Class } 0) = 0.5$ $P(\text{Class } 1) = 0.5$

Step 3: Calculate Conditional Probabilities (Likelihoods) with Laplace Smoothing

We use Laplace smoothing (add 1 to numerator, add vocabulary size to denominator) to handle words not seen in a specific class and prevent zero probabilities.

$P(\text{word} \mid \text{Class}) = (\text{Count}(\text{word in Class}) + 1) / (\text{Total words in Class} + \text{Vocabulary Size})$

$P(w_1 \mid \text{Class } 0) = (\text{Count}(w_1 \text{ in Class } 0) + 1) / (\text{Total words in Class } 0 + \text{Vocabulary Size})$
 $= (1 + 1) / (3 + 3) = 2/6 = 0.333$

$P(w_2 \mid \text{Class } 0) = (\text{Count}(w_2 \text{ in Class } 0) + 1) / (\text{Total words in Class } 0 + \text{Vocabulary Size})$
 $= (2 + 1) / (3 + 3) = 3/6 = 0.5$

$P(w_3 \mid \text{Class } 0) = (\text{Count}(w_3 \text{ in Class } 0) + 1) / (\text{Total words in Class } 0 + \text{Vocabulary Size})$
 $= (0 + 1) / (3 + 3) = 1/6 = 0.167$ (not used in this doc, but good for completeness)

$P(w_1 \mid \text{Class } 1) = (\text{Count}(w_1 \text{ in Class } 1) + 1) / (\text{Total words in Class } 1 + \text{Vocabulary Size})$
 $= (0 + 1) / (2 + 3) = 1/5 = 0.2$

$P(w_2 \mid \text{Class } 1) = (\text{Count}(w_2 \text{ in Class } 1) + 1) / (\text{Total words in Class } 1 + \text{Vocabulary Size})$
 $= (1 + 1) / (2 + 3) = 2/5 = 0.4$

$P(w_3 \mid \text{Class } 1) = (\text{Count}(w_3 \text{ in Class } 1) + 1) / (\text{Total words in Class } 1 + \text{Vocabulary Size})$
 $= (1 + 1) / (2 + 3) = 2/5 = 0.4$

Step 4: Apply Naive Bayes Formula for Class 0

We calculate the unnormalized posterior probability for Class 0 using the formula:

$$P(\text{Class } 0 \mid \text{new doc}) \propto P(\text{Class } 0) \times P(w_1 \mid \text{Class } 0) \times P(w_2 \mid \text{Class } 0) \times P(w_2 \mid \text{Class } 0)$$

$P(\text{Class } 0 \mid \text{new doc}) \propto 0.5 \times 0.333 \times 0.5 \times 0.5$
 $= 0.0416$

Step 5: Apply Naive Bayes Formula for Class 1

Similarly, we calculate the unnormalized posterior probability for Class 1:

$$P(\text{Class } 1 \mid \text{new doc}) \propto P(\text{Class } 1) \times P(w_1 \mid \text{Class } 1) \times P(w_2 \mid \text{Class } 1) \times P(w_2 \mid \text{Class } 1)$$

$P(\text{Class } 1 \mid \text{new doc}) \propto 0.5 \times 0.2 \times 0.4 \times 0.4$
 $= 0.016$

Step 6: Make Classification Decision

Compare the unnormalized posterior probabilities:

$P(\text{Class } 0 \mid \text{new doc}) = 0.0416$
 $P(\text{Class } 1 \mid \text{new doc}) = 0.016$

Since 0.0416 (for Class 0) > 0.016 (for Class 1), the new document is classified as Class 0.

Prediction: Class 0 (Negative)

Note: We only need to compare these values to find the maximum; explicit normalization (dividing by $P(\text{new doc})$) is not required for classification as it would scale both values by the same factor and not change their relative order.

Real-World Use Cases of Naive Bayes

Spam Email Filtering

Classifying emails as spam or legitimate based on word patterns and sender characteristics

Sentiment Analysis

Determining whether product reviews, social media posts, or customer feedback express positive or negative opinions

Document Categorization

Organizing news articles by topic, detecting language, or routing support tickets to appropriate departments

Why Naive Bayes excels: Fast training and prediction, scales efficiently to high-dimensional data like text, and requires relatively small training datasets to achieve good performance.

Summary & Key Takeaways

Foundation

Conditional probability is the mathematical foundation that enables Naive Bayes to make predictions

The Bridge

Bayes' theorem provides the mechanism to invert probabilities and predict class membership

The Simplification

The "naive" independence assumption makes calculations tractable for real-world problems

Practical Power

Despite mathematical simplifications, Naive Bayes performs remarkably well across diverse applications

Next Steps for Hands-On Learning

Try implementing Naive Bayes on real text datasets using Python's scikit-learn library. Start with sentiment analysis on movie reviews or spam detection on email datasets to see these principles in action.

